ERP Classification

using xDAWN + LDA

Oddball Paradigm | 10 Subjects | 32 EEG Channels

Preksha Joon
Priyanka Aravindan
Chinedu Obiefuna

NeuroHack 2026

3,600
Total Epochs

0.816
Best Accuracy

4
Models Compared

d=2.02
P300 Effect Size

The Dataset

Specifications

Paradigm: Oddball (target/non-target)

Participants: 10 subjects, 5-6 min each

Channels: 32 EEG + 1 HEOG (dropped)

Sampling: 1000 Hz, avg referenced

Trials: 60 target + 300 non-target per subject

Ratio: 1:5 class imbalance

Epoch: 750 ms ISI, [-200, 700] ms window

Challenges

- Severe 1:5 class imbalance
- Only ~60 targets per subject
- Small sample (N=10) limits generalizability

Strengths

- High efficiency: 360 trials in 5 min
- Clean design for easy epoching
- Abundant non-target data for ERP averaging

Processing Pipeline

1 Load	2 Clean	3 ICA	4 Filter	5 Epoch	6 Classify
Raw EEG 33 channels 1000 Hz	Drop HEOG Keep 32 EEG Avg-ref rank 31	1Hz copy → fit Surrogate EOG ID eye ICs	0.1–30 Hz Apply ICA Single path	[-200, 700]ms Baseline correct 3,600 epochs	4 methods LOSO eval Balanced acc

ICA uses a two-track strategy: aggressively filtered copy for training, gently filtered original for ERP preservation

Preprocessing: Why ICA?

Core Idea: ICA separates EEG into independent sources, isolating brain activity from artifacts (blinks, saccades, muscle). It cleans data ***without rejecting entire trials***, preserving statistical power for ERP analysis.

Surrogate EOG

VEOG = (Fp1+Fp2)/2
HEOG = F8 - F7
Built from EEG channels

Eye IC Detection

4 criteria voting system:
EOG correlation + topography
+ ICLabel (need 2+ votes)

Apply to Original

ICA matrix from 1Hz copy
applied to 0.1-30 Hz data
Preserves slow P300

ERP Components & Statistics

N170

Face-sensitive negative peak at posterior sites

-2.13 uV

target amplitude

Window	**130-200 ms**
Channels	**P7/P8/PO9/PO10**
Cohen's d	**-1.31**
p-value	**.003****

VPP

Vertex positive potential, frontal mirror of N170

+1.47 uV

target amplitude

Window	**130-200 ms**
Channels	**Cz/Fz/FC1/FC2**
Cohen's d	**+0.86**
p-value	**.023***

P300

Target detection response, strongest ERP effect

+4.30 uV

target amplitude

Window	**250-500 ms**
Channels	**Pz/CP1/CP2/Cz**
Cohen's d	**+2.02**
p-value	**<.001*****

Classification Methods

#	Method	Input	Key Idea	Acc
1	Traditional LDA	64 features	Mean amplitude in N170+P300 windows	0.733
2	Shrinkage LDA	192 features	Ledoit-Wolf regularization for p > n	0.754
3	Cost-Sensitive SVM	192 features	RBF + class weights + GridSearch C	0.791
4	xDAWN + LDA	Raw epochs	Supervised spatial filter maximizes ERP SNR	0.816
5	Minimum Distance to Mean (Riemann Classification)	Raw epochs	Classifies Nearest class mean on Reimann Manifold	0.602

Evaluation Strategy

LOSO (9/1)

Train on 9 subjects, test on 1. Strictest cross-subject test.

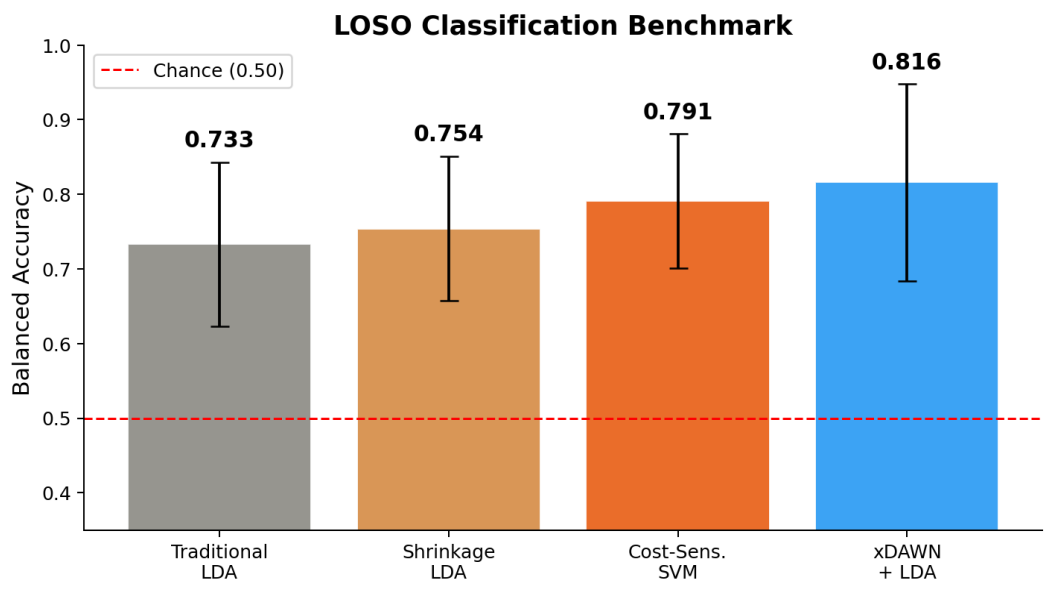
8/1/1 Split

8 train, 1 validation, 1 test. Checks model stability.

Pooled 70/10/20

Stratified per-subject split. Tests within-subject generalization.

Results: LOSO Benchmark

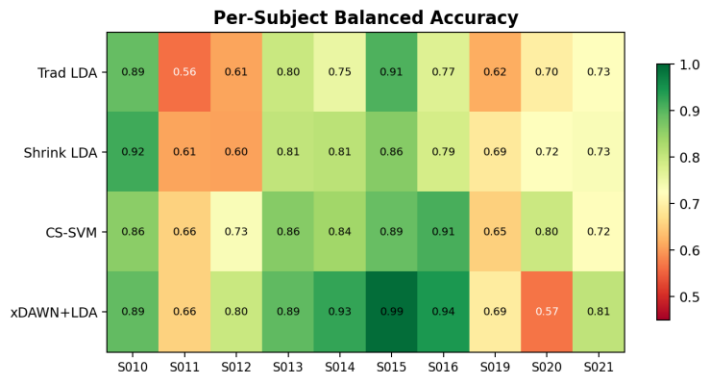


Winner

xDAWN + LDA

0.816 balanced accuracy

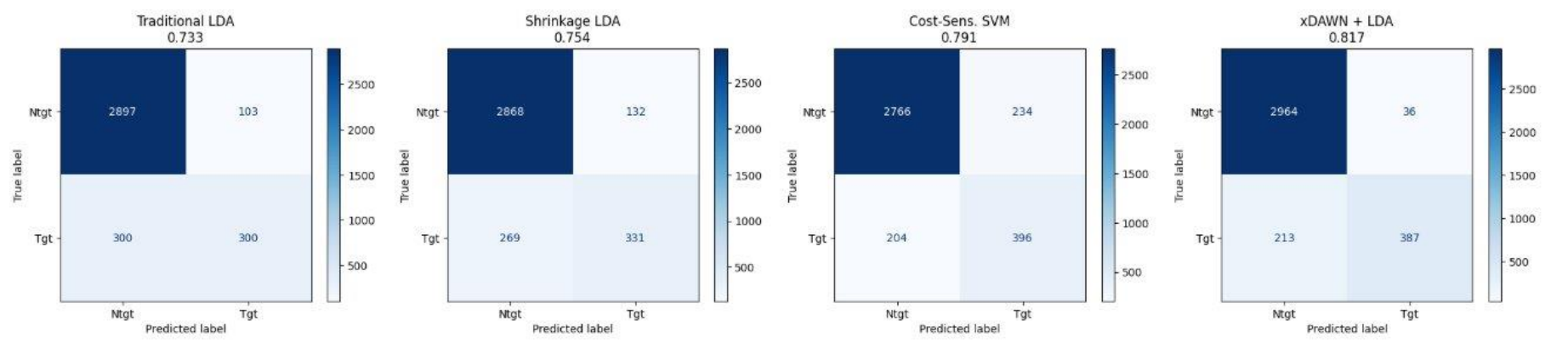
Statistically significant over Traditional LDA (p=.037)



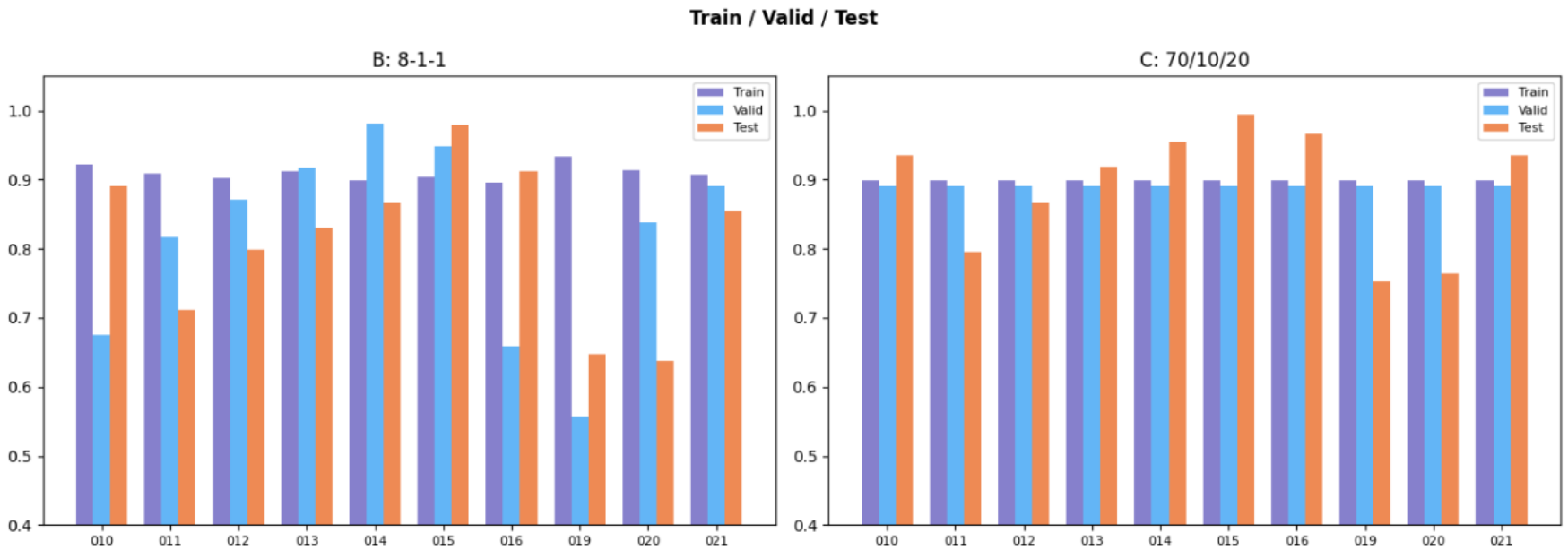
Per-subject balanced accuracy

Key: xDAWN learns optimal spatial filters from data, compressing 32 channels into 8 virtual channels that maximize target-evoked SNR. It bridges data-driven learning with regularized classification.

Results across models



Results across models



Method	Train	Valid	Test	Gap
B: 8-1-1	0.910	0.816	0.813	+0.097
C: 70/10/20	0.898	0.892	0.889	+0.010

Key Insights

xDAWN wins without domain knowledge

Never told where P300 is, yet outperforms hand-picked features. Learns optimal spatial filters directly from the data.

Subject variability is the bottleneck

Sub-015 reaches 0.992 while sub-020 drops to 0.568. A 42-point gap on the same model — the algorithm isn't the limit, inter-subject differences are.

Rich features beat basic features

Shrinkage LDA (192 features: mean+peak+variance) scores 0.754 vs Traditional LDA's 0.733 (64 features: mean only). Variance captures temporal stability.

Domain knowledge still matters

Traditional LDA (0.733) uses 60 years of ERP literature to pick exact windows. It beats any method that ignores this knowledge entirely.

ICA doesn't affect classification much

With vs without ICA: <2% difference in balanced accuracy. ICA matters for ERP amplitude statistics, not for coarse binary classification.

Future Work & Practical Implications

Future Work

- Harmonic Mean LDA for better class separation
- Channel importance analysis (which electrodes drive results)
- Advanced signal processing (wavelet, time-frequency)
- Polynomial feature expansion
- Extend to FlickerOddball paradigm
- Riemannian alignment for cross-subject transfer

Practical Implications

BCI Spellers

81.6% cross-subject with 60 trials.
Plug in, skip calibration, start typing.

Attention Monitoring

Detect rare events from a single trial
at 1:5 odds. ATC, surgical monitoring.

Clinical Screening

P300 effect (d=2.02) flags neurological
conditions. Failures are diagnostic too.

Small-Data Playbook

<20 subjects: domain features.
Spatial boost: xDAWN. Speed: Shrinkage LDA.

References

Bell, A.J. & Sejnowski, T.J. (1997). The 'Independent Components' of Natural Scenes are Edge Filters. Vision Research, 37(23), 3327-3338.

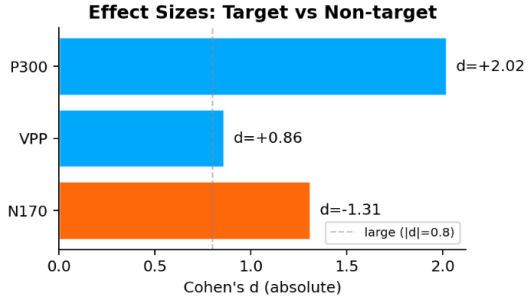
Simar, C., Petit, R., et al. (2022). Riemannian classification of single-trial surface EEG and sources during checkerboard and navigational images. PLOS ONE, 17, e0262417.

Rivet, B., Souloumiac, A., Attina, V. & Gibert, G. (2009). xDAWN Algorithm to Enhance Evoked Potentials. IEEE Trans. Biomedical Engineering, 56(8), 2035-2043.

Barachant, A., Bonnet, S., Congedo, M. & Jutten, C. (2012). Multi-class Brain-Computer Interface Classification by Riemannian Geometry. IEEE Trans. Biomedical Engineering, 59(4), 920-928.

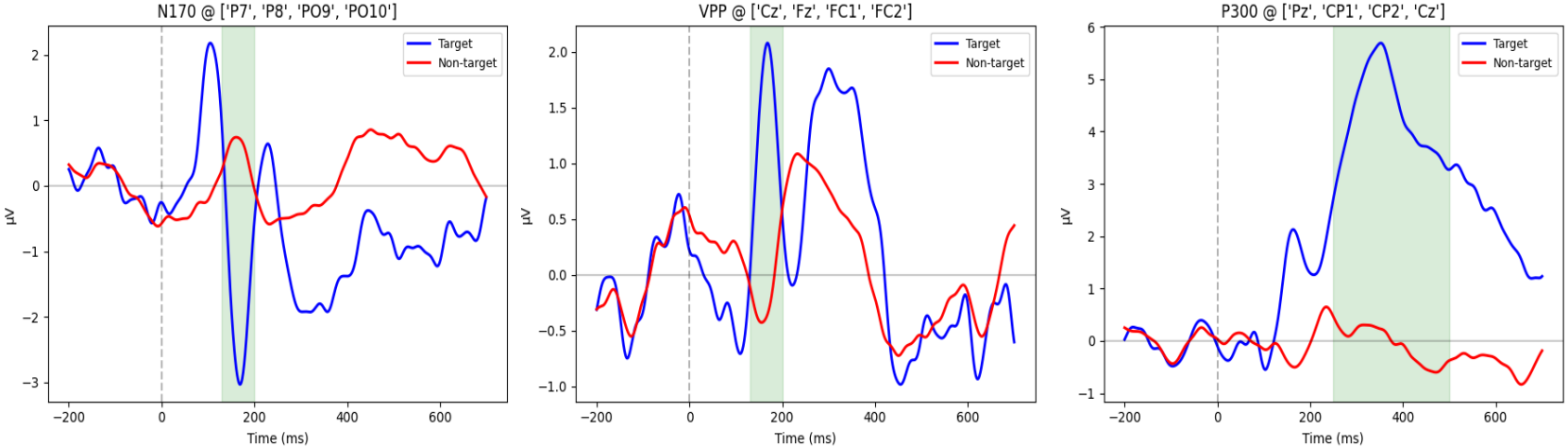
Ledoit, O. & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. Journal of Multivariate Analysis, 88(2), 365-411.

Thank you



Effect sizes for target vs non-target comparison (N=10, all $p < 0.05$)

Component	Window	Channels	Target	Non-tgt	t(9)	p	Cohen's d
N170	130-200 ms	P7/P8/PO9/PO10	-2.13 uV	+0.59 uV	-4.13	.003**	-1.31
VPP	130-200 ms	Cz/Fz/FC1/FC2	+1.47 uV	-0.19 uV	2.72	.023*	+0.86
P300	250-500 ms	Pz/CP1/CP2/Cz	+4.30 uV	-0.07 uV	6.38	<.001***	+2.02

Wilcoxon signed-rank tests confirmed all results (P300: W=0, p=.002; N170: W=3, p=.010; VPP: W=6, p=.027)

Grand Average ERPs (ICA-corrected)